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ORIGINAL ARTICLE

The Effect of Participation in the Let's Play Program on Autistic Children's Engagement and Caregiver Well-Being: A Randomized Controlled Trial

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Abstract

Purpose In Aotearoa New Zealand (NZ), families face persistent barriers to accessing evidence-based early support for Autistic children. This randomized controlled trial (RCT) evaluated the effectiveness of Let's Play; a bespoke, caregiver-mediated early support program for Autistic children and their caregivers.

Methods This single-blind (rater) RCT included 91 parent-child dyads, randomly assigned to the Let's Play program (active support; AS [$n=45$]) or a waitlist control (WLC; $n=46$). Participants were caregivers of children aged 0 to 5 years with a formal diagnosis or characteristics of autism. Let's Play was delivered over 9 weeks via group workshops and in-home coaching. Primary child and caregiver outcomes, assessed at baseline, post-support, and 6-month follow-up, included parent-child engagement and parental stress, respectively.

Results Children showed descriptive evidence of improvement in caregiver-child engagement, health-related quality of life and behaviour from baseline to post-support. Improvement in parental stress, depression and anxiety symptoms, and self-perceived parenting competence were also evident, across timepoints. However, there were no significant Group x Ttime effects for caregiver-child engagement, number of utterances or number of different words. A Group x Time effect was evident for all other child and caregiver outcome variables, underscoring the benefits of Let's Play.

Conclusions This research provides preliminary evidence of the effectiveness of a low-intensity, community-based, caregiver-mediated early support program for Autistic children's health-related quality of life and behavior and caregiver well-being. However, more targeted or sustained approaches to supporting caregiver-child engagement and vocal communication may be needed for improvement to be observed.

Clinical Trial Registration The research protocol was prospectively registered on the Australian New Zealand Clinical Trials Registry (ACTRN12622001139763).

Keywords Autism · Caregiver well-being · Caregiver-mediated support · Early support · Caregiver coaching

Introduction

Autism is a form of neurodivergence characterized by differences in social attention and communication, patterns of behavior and interests, and information processing (American Psychiatric Association [APA], 2013). Recent estimates suggest that 1 in 36 8-year-olds in the United States (Maenner et al., 2023) and 1 in 40 children in Aotearoa New Zealand (NZ) (Ministry of Health, 2023) are Autistic. This reflects a global trend of increased rates of identification of autism, attributed toward factors such as increased societal

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awareness and knowledge of autism, changes to diagnostic criteria, and greater accessibility of autism-specific supports (Zeidan et al., 2022).

While parenting an Autistic child can be a positive and rewarding experience (Voulgarakis et al., 2023), caregivers of Autistic children often experience higher parenting stress and mental health difficulties when compared to caregivers of non-Autistic children (Barroso et al., 2018; Pinquart, 2019; Shepherd et al., 2021). This can have significant secondary effects on the caregiver-child relationship, child development, and wider family well-being (Crowell et al., 2019). Existing research highlights the benefits of early autism identification and support for enhancing the health and well-being outcomes of children and their caregivers (Reichow et al., 2018; Rollins et al., 2019; Schreibman et al., 2015; Trembath et al., 2023; Weitlauf et al., 2020). Despite this compelling evidence, many NZ families face barriers to accessing diagnostic and support services. A focus on therapist-implemented, one-on-one models of delivering support, combined with a scarcity of trained professionals has resulted in inadequate and insufficient access to services for many Autistic children and their caregivers (Autism NZ, 2020).

Caregiver-implemented programs offer an alternative to clinician-delivered support. Caregiver-implemented models confer many advantages over clinician-delivered approaches as they do not depend on the availability of specialist support and have the potential to be more time- and cost-effective (Penner et al., 2015; Segal et al., 2023). Moreover, as children spend more time with their caregivers than anyone else, there are critical opportunities for embedding learning opportunities within the natural context, throughout the day (Nevill et al., 2018).

Within Developmental Relationship-Based Interventions (DRBIs), children's social relationships (especially those with their caregivers and wider family) are used as a medium for learning (Cullinane et al., 2024; Trembath et al., 2023). Children can be supported in naturalistic settings (e.g., their home or school) and during daily activities (e.g., family routines) in a way that is developmentally appropriate (Schreibman et al., 2015). Research suggests that DRBIs can have positive effects on Autistic children's engagement and communication (Binns & Oram Cardy, 2019; Boshoff et al., 2020; Deb et al., 2020; Divya et al., 2023; Sandbank et al., 2020), and caregivers' interactions with their children (Binns & Oram Cardy, 2019; Green et al., 2010; Trembath et al., 2023). However, early interventions and supports are often delivered using a one-on-one, coach-caregiver model (e.g., Abouzeid et al., 2020; Mirenda et al., 2022; Stahmer et al., 2020) which can be highly resource intensive and present the same issues as clinician-delivered support.

Group coaching offers an alternative or adjunct to one-on-one caregiver coaching, allowing coaches to assist multiple families concurrently. Further, the group context provides opportunities for caregivers to learn from one another and share their experiences (Jurek et al., 2023; Qu et al., 2022; Sengupta et al., 2023). Research into group coaching as a component of autism support highlights the feasibility and acceptability of this model (Qu et al., 2022; Sengupta et al., 2023; van Noorden et al., 2025). While there is emerging evidence of the benefits of group coaching, there are few RCTs evaluating the effects of community-based DRBIs for Autistic children and their caregivers that include group-based delivery. Further, there are few studies evaluating DRBIs that provide support for both children and their caregivers (Trembath et al., 2023).

Let's Play, a program developed by Autism NZ, is a caregiver-mediated, DRBI that is delivered using a combination of group workshops and one-on-one coaching during home visits. Autism NZ is a charitable organization. Their services include the development and delivery of programs to help educate and support Autistic people, their wider families/communities, and health and education professionals. Let's Play was developed to bridge the support gap while families await diagnosis and access to services. The program centers on the positive impact that caregivers can have on their child's development with an emphasis on caregivers creating and responding to learning opportunities that occur during play, daily activities, and routines.

This RCT aimed to evaluate the effects of the Let's Play program on: (1) caregiver-child interaction and engagement (primary child aim); (2) parental stress (primary caregiver aim); (3) caregivers' psychological distress and parenting self-efficacy (secondary caregiver aims); and (4) children's health-related quality of life and communication (secondary child aims). The social acceptability, feasibility, and cultural responsiveness of the program were also evaluated through interviews with caregivers. We hypothesized that compared to the waitlist control (WLC) condition, caregiver participation in the Let's Play program would result in (1) an increase in caregiver-child engagement and interaction; (2) a reduction in parental stress and psychological distress; (3) increased parenting self-efficacy; (4) enhanced quality of life, behavior, and communication among Autistic children; (5) maintenance of improvements in child and caregiver outcomes at follow-up; and (6) that caregivers would describe the program as socially acceptable and culturally responsive.

Methods

Design

This was a two-arm parallel group, single-blind (rater), RCT. This trial included: (a) random assignment to the Let's Play program (active support [AS]) or the WLC condition; (b) baseline assessment; (c) immediate program access (AS) or post-wait baseline assessment after a 9-week wait period, immediately followed by program access (WLC); (d) program participation (9 weeks); (e) post-support assessment; and (f) follow-up assessment (see Fig. 1). Recruitment occurred between October 2022 and June 2023.

Participants

Participating caregivers met the following inclusion criteria: (1) they had a child with an autism diagnosis or a child who had been professionally identified as having a high likelihood of autism (e.g., by a developmental pediatrician) and were awaiting further assessment; (2) their child was aged ≤ 5 years 11 months at the time of enrolment, the cut-off age for transition to formal schooling in NZ; (3) they consented to randomization; and (4) they consented to participation. Participants were excluded if: (1) they were actively participating in another parenting program; (2) their child was receiving ≥ 15 h of professional support per week; and/or (3) they did not speak sufficient English to provide fully informed consent or understand workshop content.

Randomization and Blinding

Random allocation was conducted by a biostatistician (PS) not involved with recruitment or data collection, using computer-generated balance support group randomization with a fixed block size of four. Allocation order was placed in numbered, opaque envelopes by an independent research assistant. The lead investigator (LM) was advised when to open each envelope to reveal the allocated condition by the postdoctoral fellow (AH). Within this single-blind trial, program coaches were aware of the condition each family was in. However, researchers who coded, scored, and analyzed outcomes measures were not. The blinded biostatistician (PS) conducted the main statistical analyses. The unblinded postdoctoral fellow (AH) conducted secondary analyses, safety checks via reporting on caregiver assessments, procedural integrity checks, and caregiver interviews.

Measures

Participants in each condition completed baseline, post-support, and follow-up measures. The WLC also completed measures immediately before commencing the program (post-wait baseline) to assess change during the 9-week wait period. Demographic information was collected at baseline only. Treatment acceptability was assessed at post-support only.

Primary Outcomes

Caregiver-Child Engagement

Caregiver-child engagement was coded based on 10-minute video recordings of caregiver-child play. Blind assessors used partial and whole interval measures based on Bakeman and Adamson (1984) to code 10-second intervals according to whether the child was actively engaged (person engagement and/or coordinated joint engagement) with their caregiver for ≥ 7 seconds (whole interval; Total Engagement). Child engagement was represented by the percentage of 10-second intervals in which the child demonstrated Total Engagement. Engagement states were coded into three overarching categories: Unengaged (unengaged or object-alone engagement), passive engagement (onlooking or passive joint engagement with objects or persons), and active engagement. Person engagement was coded when the child reciprocally interacted with and responded to the caregiver and exhibited awareness of the caregiver's actions (e.g., occasionally orienting face toward caregiver, following instructions). Coordinated joint engagement was coded when the child and caregiver were both engaging with the same object or play-based activity, or when the child was actively engaged with the caregiver who was interacting with the object or play-based activity in other ways (e.g., through communication, imitation, or following instructions), while alternating attention between the caregiver and the object or activity. Adapted versions of Bakeman and Adamson's (1984) framework have been utilized in a number of caregiver-mediated autism intervention evaluations, including highly powered RCTs (Kasari et al., 2010; Shih et al., 2025). Bakeman and Adamson's coding methods have demonstrated high inter-rater reliability (e.g., intraclass correlation of 0.94; Nordahl-Hansen et al., 2016).

Parental Distress

The Parent Distress subscale of the Parenting Stress Index (Fourth Edition, Short Form; PSI-4-SF) was used to assess parenting distress (primary caregiver outcome). The Parental Distress subscale has good reliability (Cronbach's $\alpha = 0.88$) based on self-report by caregivers of Autistic children (Zaidman-Zait et al., 2010) and the PSI-F has been widely used in past research with parents of Autistic children (Ekas et al., 2023; Liu et al., 2023).

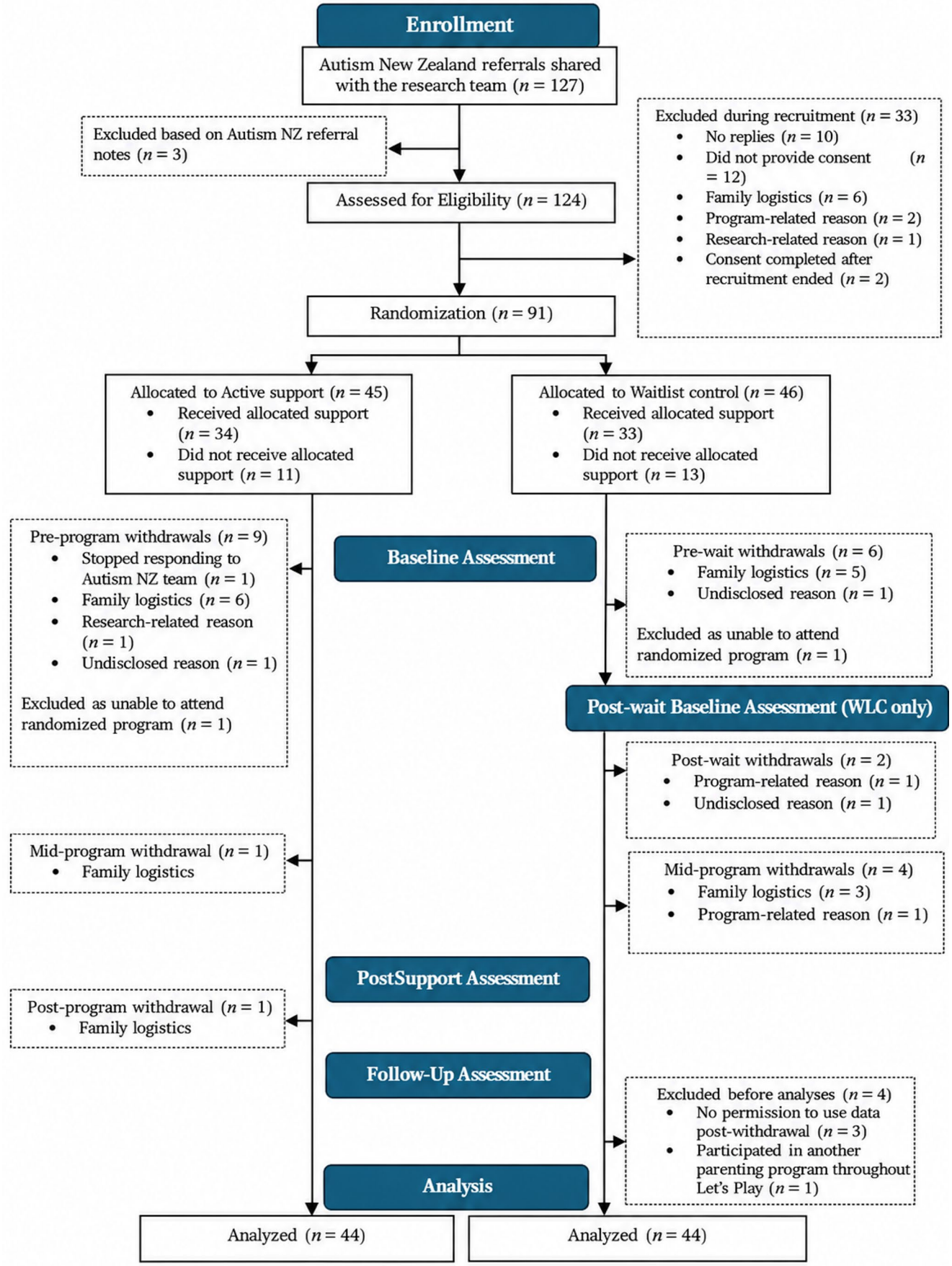

Enrollment
Autism New Zealand referrals shared with the research team (n = 127)
Excluded based on Autism NZ referral notes (n = 3)
Assessed for Eligibility (n = 124)
Excluded during recruitment (n = 33)
No replies (n = 10)
Did not provide consent (n = 12)
Family logistics (n = 6)
Program-related reason (n = 2)
Research-related reason (n = 1)
Consent completed after recruitment ended (n = 2)
Randomization (n = 91)
Allocated to Active support (n = 45)
Received allocated support (n = 34)
Did not receive allocated support (n = 11)
Allocated to Waitlist control (n = 46)
Received allocated support (n = 33)
Did not receive allocated support (n = 13)
Pre-program withdrawals (n = 9)
Stopped responding to Autism NZ team (n = 1)
Family logistics (n = 6)
Research-related reason (n = 1)
Undisclosed reason (n = 1)
Excluded as unable to attend randomized program (n = 1)
Baseline Assessment
Pre-wait withdrawals (n = 6)
Family logistics (n = 5)
Undisclosed reason (n = 1)
Excluded as unable to attend randomized program (n = 1)
Post-wait Baseline Assessment (WLC only)
Post-wait withdrawals (n = 2)
Program-related reason (n = 1)
Undisclosed reason (n = 1)
Mid-program withdrawal (n = 1)
Family logistics
Mid-program withdrawals (n = 4)
Family logistics (n = 3)
Program-related reason (n = 1)
PostSupport Assessment
Post-program withdrawal (n = 1)
Family logistics
Follow-Up Assessment
Excluded before analyses (n = 4)
No permission to use data post-withdrawal (n = 3)
Participated in another parenting program throughout Let's Play (n = 1)
Analysis
Analyzed (n = 44)
Analyzed (n = 44)

Fig. 1 CONSORT Let's play participant flow diagram

Secondary Child Outcomes

Number of Total Utterances and Total Different Words

Ten-minute video samples were analyzed using the Systematic Analysis of Language Transcripts, New Zealand/Australia (SALT-NZAU 16; Miller et al., 2015) to assess children's number of total utterances (i.e., number of unintelligible/intelligible sentences spoken by the child; represented by the number of lines of child dialogue in the coder's transcript) and total different words. The SALT-NZAU software has been used in past research to measure Autistic children's language use (Hampton et al., 2020; Hao et al., 2021).

Health-Related Quality of Life

The Pediatric Quality of Life Inventory Generic Core Scales (PedsQL 4.0; Varni et al., 2001; Varni et al., 1999) provides a measure of children's health-related quality of life. Total scores were used for this analysis.

Behavior

The Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997) provided a measure of child behavior. The 25-item, caregiver-report measure includes five subscales: Emotional Problems, Conduct Problems, Hyperactivity, Peer Problems, and Prosocial Behavior. Total Difficulties scores (total ratings on the Emotional Problems, Conduct Problems, Hyperactivity, and Peer Problems subscales) were used in this analysis. The SDQ is commonly used in autism research (e.g., Del Bianco et al., 2024; Wright et al., 2023) and total scores have acceptable reliability (D'Souza et al., 2017; Thompson et al., 2021).

Secondary Caregiver Outcomes

Caregiver Mental Health Symptoms

The Depression, Anxiety, and Stress Scale-21 (DASS-21; Henry & Crawford, 2005) is a 21-item self-report measure of symptoms of depression, anxiety, and stress (Henry & Crawford, 2005), with separate subscales associated with each state. Only caregivers' Depression and Anxiety subscale scores were included in the final analyses as caregiver stress is captured using the PSI-4-SF. Each of these subscale scores have good reliability (Henry & Crawford, 2005).

Parenting Sense of Competence

The Parenting Sense of Competence Scale (PSOC; Johnston & Mash, 1989) provided a caregiver-reported measure of parenting self-competence across 16-items. Two subscales are embedded in the measure (Parental Satisfaction and Parental Efficacy). Overall perceived parenting competence score (total ratings on the measure; possible range: 16–96) was used in the final analysis. These scores have acceptable reliability (Johnston & Mash, 1989).

Process Measures

Treatment Acceptability

The Treatment Acceptability Rating Form-Revised (TARF-R; Reimers et al. 1992a) is a 20-item parent-report measure that was completed post-support as a measure of caregiver-perceived program acceptability. Ratings were provided across six subscales (Effectiveness, Reasonableness, Willingness, Cost, Negative Side-effects, and Disruption/Time). The Negative Side-effects and Disruption/Time subscales, ratings were summed to provide a total treatment acceptability score. The TARF-R has adequate internal consistency (α=0.69–0.95; Reimers et al. 1992a, 1992b) and is widely to measure the acceptability of autism supports (e.g., Abouzeid et al., 2020; Waddington et al., 2020). In addition to the TARF-R, 16 caregivers (18%) agreed to be interviewed to determine the social and cultural acceptability of the program (e.g., "Did you and your whānau [family] find the program culturally safe, appropriate and engaging?" and "Do you think there is anything that could be changed to ensure it is more culturally responsive?"). Priority was given to Māori and Pacific participants, consistent with the focus on understanding cultural appropriateness. All interviews were conducted by a member of the research team who was not involved in programme delivery. The interview guide is included as supplementary materials (S1).

Session Attendance

Program coaches recorded the number of workshops and in-home coaching sessions caregivers attended as a measure of support dosage.

Therapist Procedural Integrity

Observers from the research team developed a bespoke 15-item group workshop checklist and an 11-item home visit checklist to measure fidelity of program implementation. Total Therapist Procedural Integrity (TPI) was calculated by dividing the number of checklist items observed by

the total number of checklist items, multiplied by 100. Reliability of fidelity recording was determined by co-coding group workshops and home visits. Interobserver Agreement (IOA) was calculated using Intraclass Correlation Coefficients (ICCs; two-way random effects model, absolute agreement, single rater; Koo & Li, 2016).

Adverse Events

A log was used to record adverse events, including safety checks with caregivers scoring highly on the stress and mental health measures (PSI-4-SF and DASS-21, respectively) and concerns reported by participants.

Procedures

Ethical approval was obtained from the New Zealand Health and Disability Ethics Committee (Ref #13041). The research protocol was prospectively registered on the Australian New Zealand Clinical Trials Registry (ACTRN12622001139763) and subsequently published (Hinten et al., 2025). This protocol followed the CONSORT (Consolidated Standards of Reporting Trials) statement for randomized trials (Schulz et al., 2010). Referrals for the Let's Play program were received from professionals in Auckland, NZ. Referred families were screened by Autism NZ (program provider) against inclusion criteria and subsequently provided with information about the research evaluation. Interested families were referred to the research team, and research eligibility screening and consent processes were undertaken. Following consent, participants were randomly assigned to the AS or WLC condition. Those in AS received immediate access to the next available program, while those in the WLC were able to access the program following a 9-week wait period.

The Let's Play Program

Let's Play was developed and delivered by Autism NZ staff who are qualified specialist teachers and/or parents of Autistic individuals. Educators and parent coaches received training delivered by Autism NZ, through standard training mechanisms, including observation of delivery, workshops, modeling and practice. Educators were qualified teachers and early intervention specialist teachers. Parent coaches had lived experience of parenting Autistic children, enabling a peer support approach that facilitated the reciprocal exchange of experiences and knowledge, increased social connections, and reduced feelings of isolation. They offered insights and provided new ways for other whānau to build knowledge. The program was delivered over 9 weeks and included three, 3-hour group workshops (6-10 families per group; Weeks 1, 3, and 5) delivered in-person in a community clinic, and five, 1-hour, in-home, one-on-one coaching sessions (Weeks 2, 4, and 6-8). Families identified one family member to attend all sessions. Let's Play is based on the principles of Joint Action Routines (Ratner & Bruner,1978), Vygotsky's (1978) Zone of Proximal Development, guided participation (Rogoff, 2003), Rush and Shelden's (2020) adult learning and coaching models, the Māori (Indigenous people of NZ) practice of tuākana-tēina (i.e., coaching of novice peers by experienced peers), and parental attunement (i.e., building an emotional relationship through appropriately translating and responding to their child's cues during play and joint activities; Schore, 2001). Let's Play can be considered a DRBI (Cullinane et al., 2024). The program targeted three domains: (1) caregiver play, (2) caregiver communication techniques, and (3) family well-being. The program aimed to provide caregivers with a range of strategies to apply during naturally occurring, everyday routines, to support caregiver-child engagement, shared attention, social reciprocity, and their child's imitation, emotional regulation, problem solving, and communication. Each group workshop focused on a different aspect of play (i.e., the basics of play, following your child's lead, and using everyday routines). Content was delivered within presentations describing the theoretical underpinnings of the focus area, examples of application of techniques, video exemplars, and group discussions. Printed handbooks were provided that included workshop content and homework activities. During in-home sessions, coaches provided coaching as caregivers implemented strategies taught during workshops. Feedback was provided synchronously or based on recorded video. Caregiver successes and challenges were also discussed. The framework for program delivery is published in the study protocol (Hinten et al., 2025). Those in the WLC continued to access any usual support while waiting to start Let's Play.

Power Estimation

A power calculation was used to estimate minimum participants numbers to identify an effect on caregiver-child engagement (primary child outcome) if one existed in the data. The power calculation suggested a minimum requirement of 26 families per condition to find a medium effect size (Cohen's $d = 0.80$; based on Kasari et al., 2010) with 80% power. Recruiting 32 families per condition allowed for 20% attrition.

Statistical Analysis

The data analysis plan was registered on the Open Science Framework prior to commencing data analysis: https://osf

.io/q6kbu/. Based on an Intention-to-Treat (ITT) analysis, submitted data were used from all participants regardless of whether they completed the entire Let's Play program (i.e., attended all group workshops and home visits). The main analysis was completed in Stata SE (version 18.0). Student's *t*-tests and Fisher's exact tests were used to compare continuous and categorical variables at baseline. Recognizing that regression-to-the-mean is a ubiquitous phenomenon in repeated data and should always be considered as a possible cause of an observed change, the analysis of covariance (ANCOVA) approach recommended by Barnett et al. (2005) was adopted for all primary analyses comparing AS and WLC participants. Secondary analyses were completed in SPSS (version 29.0.0.0). This included Pearson's correlations to analyze the relationships between participants' scores on the various outcome measures, and the intraclass correlations (ICCs) for interobserver agreement (IOA). Significance was two-sided and set at $\alpha = 0.05$. Reflexive thematic analysis was used to analyze themes from interview notes, based on the six steps outlined by Braun and Clarke (2019). ANCOVAs were undertaken to compare group differences between the WLC and AS between time-point pairs. The first time-point pairs compared change from baseline to post-support among the AS condition, with change in post-wait baseline to post-support for WLC. Post-support to follow-up was the second time-point pair compared for both the AS and WLC condition. When no significant differences emerged between groups, estimated change across time-points was also analyzed for the combined conditions rather than for each group separately. This approach is useful for estimating the overall change over time associated with the intervention, but it ignores the WLC feature of the study design. It offers descriptive evidence of improvement, but unlike the ANCOVA analysis, it cannot be used to determine program effectiveness. When significant differences emerged (e.g., Depression Symptoms in caregivers), then both the group-specific and overall effect sizes were presented. For between-group comparisons, WLC participants' post-wait baseline measurements were compared to the AS participants' post-support measurements (end of their 9 weeks in the Let's Play program) to compare AS participants' scores after they have been in the Let's Play program to the WLC participants' scores when accessing care-as-usual over the same duration.

Results

Baseline Comparisons

A total of 124 families met eligibility criteria for the study and of those, 91 provided consent and were randomized to the AS ($n = 45$) or WLC ($n = 46$) condition (see Fig. 1). Two families were excluded from the analyses either because they did not meet the study inclusion criteria ($n = 1$; WLC) or could not attend the workshop for the condition they were randomized into ($n = 2$; 1 per condition). Hence, 44 families remained in each condition.

Participant demographics including child diagnostic status and child and caregiver age and ethnicity are presented in Table 1. Baseline measure score comparisons between AS and WLC participants, and correlations between baseline scores, are presented in Supplementary Tables S2 and S3, respectively. These data suggest that there were no significant differences between conditions for the child and caregiver demographic characteristics and no significant baseline differences between conditions across any child or caregiver outcome measures.

Change Over Time-Points

Descriptive data and change in primary and secondary child and caregiver outcomes across time-points are presented in Table 2. Mean change refers to change in mean scores across time-points for the combined group data analysis (i.e., AS + WLC; see data analysis section).

Child Outcome Measures

Children showed significant improvement from baseline (i.e., baseline [AS] and post-wait baseline [WLC]) to post-support on caregiver-child engagement (primary outcome; $M = 12.8$; 95% Confidence Interval [CI] = 3.6, 21.9, $p \leq 0.01$), health-related quality of life ($M = 2.7$; 95% CI = 0.1–5.2, $p \leq 0.05$), and behavior ($M = -1.9$; 95% CI = −3.0 – −0.9, $p \leq 0.001$). They showed no significant changes in the number of utterances and different words. Between post-support and follow-up, participants showed significant improvement in number of utterances ($M = 10.4$; 95% CI = 4.3–16.4, $p \leq 0.01$), and number of different words ($M = 4.3$; 95% CI = 1.6–6.9, $p \leq 0.01$). They showed no significant further change in behavior, engagement, or health related quality of life.

Caregiver Outcome Measures

Caregivers showed significant improvement across time-points on all outcomes, including parental distress ($M = -4.2$; 95% CI = −5.9 – −2.4, $p \leq 0.001$), depression symptoms ($M = -1.4$; 95% CI = −2.2 – −0.5, $p \leq 0.01$), anxiety symptoms ($M = -0.9$; 95% CI = −1.6 – −0.2, $p < 0.01$), and self-perceived parenting competence ($M = 6.3$; 95% CI = 3.7–8.8, $p \leq 0.001$). Significant improvement was demonstrated between post-support and follow-up for

Table 1 Participant child and caregiver demographic information

Variable	Mean (SD)/Count (%)			
	Total sample ($n=88$)	Active support ($n=44$)	Waitlist control ($n=44$)	Fisher's exact p value or t-test value
Mean (*SD*) child age at consent (months)	38.8 (9.3)	39.4 (9.0)	38.2 (9.6)	$t(83)=0.59$
Child sex				$p=0.78$
Male	69 (78)	36 (82)	33 (75)	
Female	15 (17)	7 (16)	8 (18)	
Child ethnicity*				$p=0.69$
Asian	24 (27)	14 (32)	10 (23)	
Pacific	20 (23)	8 (18)	12 (27)	
European	18 (20)	8 (18)	10 (23)	
Māori	12 (14)	7 (16)	5 (11)	
Other	7 (8)	5 (11)	2 (5)	
Child's diagnostic status				$p=0.71$
Formal autism diagnosis	62 (70)	32 (73)	30 (68)	
Awaiting diagnosis	8 (9)	5 (11)	3 (7)	
Mean (*SD*) caregiver age at consent (years)	35.7 (6.7)	36.4 (7.4)	35.1 (5.9)	$t(67)=0.77$
Caregiver sex				$p=0.99$
Female	75 (85)	39 (89)	36 (82)	
Male	10 (11)	5 (11)	5 (11)	
Caregiver ethnicity*				$p=0.23$
Asian	21 (24)	14 (32)	7 (16)	
European	19 (22)	9 (20)	10 (23)	
Pacific	18 (20)	6 (14)	12 (27)	
Māori	5 (6)	3 (7)	2 (5)	
Other	5 (6)	2 (5)	3 (7)	
Caregiver's relation to child				$p=0.65$
Mother	72 (82)	36 (82)	36 (82)	
Father	10 (11)	5 (11)	5 (11)	
Grandmother	2 (2)	2 (5)	0 (0)	
Aunt	1 (1)	1 (2)	0 (0)	

n = participant count; *SD* = Standard Deviation. *Ethnicities were priority coded if multiple were provided: Māori > Pacific > Asian > Middle Eastern, Latin American, and African (MELAA) > European. For example, if a participant listed that they are of Māori, Pacific, and European descent, their ethnicity would be coded as the earliest entry on this list (i.e., Māori). There was one set of twins in the study. As only one twin was used in the analyses (randomly chosen), only one twin is represented here. Missing data - Child's age: Waitlist Control (WLC) $n = 3$; Child sex: Active Support (AS) $n = 1$, WLC $n = 3$; Child ethnicity: AS $n = 2$, WLC $n = 5$; Child's diagnostic status: AS $n = 7$, WLC $n = 11$; Caregiver age: AS $n = 10$, WLC $n = 9$; Caregiver sex WLC $n = 3$; Caregiver ethnicity: AS $n = 10$, WLC $n = 10$; Caregiver's relation to child WLC $n = 3$)

self-perceived parenting competence ($M = -3.3$; 95% CI $= -6.4 - -0.3$, $p < 0.05$). However, no further significant changes were found in parental distress or depression and anxiety symptoms.

Between-Group Comparisons

Descriptive statistics and ANCOVA outputs are presented in Table 3 for each measure and across conditions. These data compare post-support (AS) with post-wait baseline (WLC; see data analysis section).

Primary Outcomes

There was no significant difference between groups or Group × Time effect for caregiver-child engagement (F [2, 54]$=13.16$) suggesting change in caregiver-child engagement was not related to program participation. However, a significant between-group difference and Group × Time interaction was evident for parental distress (F [2, 61]$=59.23$, $p \leq 0.001$) suggesting that those that received support experienced greater reduction in parental distress than those that received care-as-usual over the equivalent 9-week period.

Secondary Outcomes

There was no significant difference between groups or Group × Time effect for number of utterances (F [2, 54]$=53.64$) or number of different words (F [2, 54]$=55.96$). However, a significant between-group difference and Group × Time effect was evident for health-related quality of life (F [2, 61]$=21.56$, $p \leq 0.05$) and behavior (F [2, 61]$=59.23$, $p \leq 0.05$). There was a significant between group difference and Group × Time effect for all secondary caregiver outcomes including depression (F [2, 61]$=17.59$, $p \leq 0.01$) and anxiety (F [2, 61]$=19.27$, $p \leq 0.01$) symptoms and parenting sense of competence (F [2, 61]$=23.77$, $p \leq 0.001$). These data suggest that participants showed significantly greater improvement in health-related quality of life, behavior, and caregiver symptoms of depression, anxiety, and sense of competence, compared to those that received care-as-usual over the same 9-week period. However, such effects were not evident for child communication outcomes.

Treatment Acceptability Ratings

Participants' ratings of the acceptability of the Let's Play program are shown in Table 4. Participants rated that they understood the program well (Understanding $M=6.7$, $SD=0.6$) and acceptability was high on all acceptability components measures (i.e., Effectiveness, Reasonableness, Willingness, Cost, and Negative Side-effects). The dimension receiving the lowest rating proportional to the maximum possible score was Disruption/Time ($M=14.4$, $SD=3.0$).

Table 2 Descriptive statistics and score changes for the child and caregiver outcome measures

Outcome				Baseline to Post-support		Post-support to Follow-up	
	Baseline	Post-support	Follow-up	Grp diff.	Mean Change	Grp diff.	Mean change
	Mean (SD)	Mean (SD)	Mean (SD)	p-value	(95% CI)	p-value	(95% CI)
Child Outcome Measures							
Caregiver child engagement				0.99	12.8 (3.6, 21.9)**	0.24	7.0 (−1.1, 15.1)
Active support	40.3 (33.2)	53.0 (29.1)	56.5 (31.0)				
Waitlist control	49.5 (35.3)	57.2 (33.4)	69.7 (33.2)				
Number of utterances				0.32	2.1 (−1.9, 6.1)	0.63	10.4 (4.3, 16.4)**
Active support	16.3 (20.3)	18.0 (23.4)	26.3 (30.7)				
Waitlist control	22.6 (27.0)	19.5 (25.4)	32.3 (35.0)				
Number of different words				0.94	1.5 (−0.5, 3.4)	0.85	4.3 (1.6, 6.9)**
Active support	6.6 (9.3)	7.7 (13.1)	12.6 (18.3)				
Waitlist control	10.6 (14.8)	10.7 (15.5)	15.0 (19.3)				
Health-related quality of life				0.46	2.7 (0.1, 5.2)*	0.95	−0.5 (−3.9, 2.8)
Active support	64.6 (13.2)	68.4 (10.5)	66.8 (15.4)				
Waitlist control	63.3 (11.5)	66.1 (9.6)	64.7 (12.5)				
Behavior differences†				0.42	−1.9 (−3.0, −0.9)***	0.75	0.9 (−0.5, 2.3)
Active support	17.5 (6.1)	15.1 (4.9)	16.5 (5.8)				
Waitlist control	17.4 (5.9)	15.5 (4.9)	16.9 (5.7)				
Caregiver Outcome Measures							
Parental distress†				0.33	−4.2 (−5.9, −2.4)***	0.24	1.2 (−0.5, 2.8)
Active support	32.9 (8.3)	27.6 (8.0)	31.2 (7.6)				
Waitlist control	31.9 (7.4)	29.2 (6.7)	29.9 (7.8)				
Depression symptoms†				0.35	−1.4 (−2.2, −0.5)**	0.03	0.9 (−0.1, 1.9)
Active support	4.1 (3.1)	2.3 (2.1)	4.2 (4.2)				2.2 (0.4, 4.0)*
Waitlist control	3.7 (3.2)	2.7 (2.3)	2.3 (2.3)				−0.3 (−1.3, 0.7)
Anxiety symptoms†				0.48	−0.9 (−1.6, −0.2)**	0.14	0.5 (−0.2, 1.3)
Active support	3.0 (2.9)	1.8 (2.0)	2.9 (2.7)				
Waitlist control	3.1 (2.6)	2.1 (2.4)	2.0 (2.1)				
Parenting sense of competence				0.74	6.3 (3.7, 8.8)***	0.65	−2.7 (−5.1, −0.4)*
Active support	62.4 (10.1)	68.8 (10.6)	67.0 (11.7)				
Waitlist control	62.3 (8.7)	69.6 (10.4)	66.7 (10.0)				

Significance asterisk system: *p<0.05; **p<0.01; and ***p<0.001. †Lower scores on these measures indicate better mental health

Table 3 Support effects for child and caregiver outcome measures comparing WLC Post-Wait baseline scores to AS Post-Support scores

Outcome measure	Possible score	Waitlist Control Post-Wait Baseline scores		Active Support Post-Support scores		
	Ranges	*n*	mean (SD)	*n*	mean (SD)	Mean diff. (95% CI)
Child Outcome Measures						
Caregiver-child engagement	0–100	31	49.5 (35.3)	27	53.0 (29.1)	14.2 (−0.4, 28.9)
Number of utterances	0+	31	22.6 (27.0)	27	18.0 (23.4)	3.4 (−4.7, 11.6)
Number of different words	0+	31	10.6 (14.8)	27	7.7 (13.1)	−0.1 (−4.5, 4.3)
Health-related quality of life	0–100	33	63.3 (11.5)	31	68.4 (10.5)	4.5 (0.1, 8.8)*
Behavior differences	0–40	33	17.4 (5.9)	31	15.1 (4.9)	−2.0 (−3.6, −0.4)*
Caregiver Outcome Measures						
Parental distress	12–60	33	31.9 (7.4)	31	27.6 (8.0)	−5.1 (−7.9, −2.4)***
Depression symptoms	0–21	33	3.7 (3.2)	31	2.3 (2.1)	−1.5 (−2.6, −0.4)**
Anxiety symptoms	0–21	33	3.1 (2.6)	31	1.8 (2.0)	−1.3 (−2.2, −0.4)**
Parenting sense of competence	16–96	33	62.3 (8.7)	31	68.8 (10.6)	6.4 (2.6, 10.3)***

n = participant count; SD = Standard Deviation; CI = Confidence Interval. Significance asterisk system: *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001

Table 4 Descriptive statistics for treatment acceptability ratings

TARF-*R* measure	Mean score (SD)			
	Possible score range	Whole sample ($n=58$)	Active support ($n=30$)	Waitlist control ($n=28$)
Total acceptability score	17–119	103.3 (10.7)	103.0 (11.6)	103.6 (9.9)
Effectiveness	3–21	18.9 (2.6)	18.9 (2.8)	18.9 (2.4)
Reasonableness	3–21	19.6 (2.1)	19.6 (2.3)	19.5 (2.0)
Willingness	3–21	18.8 (2.0)	18.5 (2.0)	19.2 (2.0)
Cost	2–14	12.6 (2.5)	12.8 (2.5)	12.4 (2.4)
Negative side-effects*	3–21	19.0 (2.8)	19.0 (3.1)	18.9 (2.5)
Disruption/Time*	3–21	14.4 (3.0)	14.2 (3.1)	14.6 (2.9)
Severity	2–14	8.0 (2.8)	8.1 (3.1)	7.8 (2.6)
Understanding	1–7	6.7 (0.6)	6.7 (0.7)	6.6 (0.6)

n = participant count; *SD* = Standard Deviation. TARF-R = Treatment Acceptability Rating Form – Revised. *The scores on these two subscales are reverse-scored so that higher scores indicate higher acceptability. There were no significant between-condition differences in the ratings (i.e., no *t*-test *p* value was < 0.05)

Therapist Procedural Integrity

Independent raters completed a TPI checklist for 5/30 group workshops (17%) and 30/193 home visits (16%). For all five workshop sessions, TPI was 100%. Over the 30 observed home visits, mean TPI was 83% ($SD=13.0$; range: 50–100). The following three items were the most missed; 'The caregiver was shown the video uninterrupted' ($n=14$), 'They collaborated on the Planning Sheet' ($n=10$), and 'A summary of the planning was provided as the visit drew to a close' ($n=8$). Of the 30 observed home visits, 23 were coded by just one rater, and 7 (23%) were observed by 2 raters. The interobserver agreement suggested good reliability (ICC = 0.85; Koo & Li, 2016).

Language Coding Interobserver Agreement

A total of 201 submitted videos met research requirements for analysis (e.g., at least 5 min of continuous recording; no un-consenting people were present in the videos). To check IOA for the SALT-NZAU, 20% of these videos ($n=41$) were coded by both language coders. ICCs were calculated separately for the number of utterances and number of different words measures. IOA for the number of different words suggested good reliability (ICC = 0.81), while the number of utterances IOA suggested only moderate reliability (ICC = 0.54).

Session Attendance

Most participants completed the entire Let's Play program, attending all group workshops (AS $n=34$, 77%; WLC $n=30$, 68%) and home visits (AS $n=25$, 57%; WLC $n=23$, 52%). Further, it was more common for participants to miss only one home visit (AS $n=6$, 14%; WLC $n=5$, 11%) rather than two (AS $n=3$, 7%; WLC $n=1$, 2%), three (AS $n=0$; WLC $n=3$, 7%), or four (AS $n=1$, 2%; WLC $n=2$, 5%). The subset of participants attending no group workshops or home visits are those who withdrew from Let's Play and the evaluation study before they began the program (AS $n=9$; WLC $n=6$). Attendance rates did not differ significantly between conditions (Fisher's exact test: group workshop $p=0.32$; home visits $p=0.50$).

Adverse Events

A small proportion of caregivers ($n=7$) reported concerns related to data collection procedures, including difficulties recording caregiver-child interactions for 10 continuous minutes ($n=4$) and technical difficulties collecting and uploading quality recordings. Three caregivers also reported that the interaction task felt contrived. Additional concerns related to time demands and competing priorities (e.g., arranging childcare to attend sessions; $n=4$), and challenges with communication from program staff ($n=2$).

Qualitative Interview Data

Sixteen caregivers completed interviews (one New Zealand European [NZE], three Māori, six Asian, and seven Pacific Peoples). Several themes emerged from the data. Reportedly, participation in the Let's Play program increased caregiver knowledge and skills, helped in connecting with others, and enhanced caregiver-child relationships. These are described in full below.

Increasing Knowledge and Skills

Caregivers valued the increased autism knowledge and practical skills they gained by participating in the Let's Play program. The fusion of theoretical group workshop sessions with practical and idiosyncratic home visits was valued. One caregiver described this approach as "theory plus practical." Caregivers' knowledge of play changed because of their participation which in turn changed how they approached play. For example, caregivers said play can happen "anytime and anywhere" and that it was important to focus on "quality of play over the quantity." This contradicted their earlier beliefs that their child would play alone if left to their own devices, or that caregivers needed to carve out specific times to play. Caregivers also expressed that their enhanced play skills led to better communication, engagement, and understanding in their caregiver-child relationship, as well as increased confidence as parents. These caregiver

reflections imply that a theoretical base that emphasizes the importance of play, an understanding of what play is, and practical play implementation suggestions is perceived as effective for improving both child and parent outcomes.

Connecting With Others

Caregivers shared that they had experienced loneliness due to their child's autism diagnosis. For example, some had narrow support networks because their family, friends, or wider community did not understand autism. By participating in Let's Play they met other families who "understood [their] struggles" as they were "surrounded by parents who get it, children who do similar things." Meeting other families helped ease caregivers' feelings of social isolation. Although the caregivers were from a range of backgrounds, they could "resonate" with the stories and experiences other caregivers shared in the group workshop sessions, and they could offer support to each other. Families reported feeling committed to the program and having good relationships with each other, allowing friendships to develop over the course of the program. For example, some caregivers mentioned that, after the program ended, they arranged play dates with families they had befriended during the program. Therefore, participating in Let's Play enabled caregivers to feel connected to a community of people with shared experiences and to create broader support networks.

Enhancing Caregiver-Child Relationships

Caregivers all noticed positive changes in their caregiver-child relationships throughout their time in Let's Play. They spoke of identifying their child's strengths and recognizing the cues their child used to communicate. One caregiver said, "We changed the way we did things which resulted in a positive change of behavior." Implementing their newly acquired skills had led to increased caregiver-child engagement through increased direct communication (verbal and non-verbal) and initiation of interactions by their child. For example, one father said that his child rarely interacted with him, but post-support his child "initiates play and brings items to me to indicate what they need." Developing new skills had multiple positive changes for caregivers, including increasing their confidence, reducing their anxiety, aligning their expectations for their child, and helping them teach their child how to navigate their daily life (e.g., potty training).

Cultural Responsiveness of the Program

Caregivers described the program as culturally responsive and inclusive and none made suggestions for improvement. Individuals reported that the program catered to a range of cultures and valued the participation of caregivers from different cultural backgrounds within workshops, identifying this as an important learning opportunity. Individuals further valued the inclusion of Te Reo (Māori language) and tikanga (Māori cultural practices and protocols) which were incorporated through practices such as opening and closing workshops with karakia (prayer/incantation or blessing). Individual responses underscored different cultural interpretations of characteristics of autism (e.g., in some cultures, children are taught not to make eye contact with elders as a sign of respect, and as such, this may not be a characteristic of autism), intergenerational differences in understandings and acceptance of autism, and the need to adapt culturally embedded parenting practices; though these were not considered negative aspects of the program. Finally, the presentation of all content in English was identified by one parent as a barrier to participation for non-English speaking caregivers.

Discussion

To our knowledge, this is the first RCT to evaluate the effectiveness of a community-based, hybrid (group workshops plus individual coaching) model of early support designed to enhance caregiver-child engagement and well-being among Autistic children and their caregivers. Results demonstrated significant short-term improvement in many child and caregiver outcomes, with qualitative data suggesting that increased caregiver knowledge, confidence, and social connection may underlie these improvements.

Child participants showed significant improvements from baseline to post-support in caregiver-child engagement, health-related quality of life, and behavior. However, no significant Group x Time effects were found for caregiver–child engagement. This suggests that, although engagement improved over time, these changes cannot be attributed to participation in the Let's Play program and may instead reflect natural developmental progression or other external influences. This distinction is important, as prior research has demonstrated that that caregiver-mediated, naturalistic, and play-based approaches can enhance caregiver-child engagement and interaction (Binns & Oram Cardy, 2019; Green et al., 2010; Trembath et al., 2023; Waddington et al., 2021). However, the current findings do not provide evidence of a program-specific effect on this primary outcome.

In contrast, improvements in child behavior and health-related quality of life were supported by evidence of significant Group x Time effects, suggesting that these gains are associated with participation in Let's Play, underscoring the broader, secondary benefits of this model of support.

Interestingly, no significant changes were observed from baseline to post-support for children's numbers of utterances and different words. However, significant improvement emerged at follow-up. Consistent with existing research (Ingersoll et al., 2024; Pickles et al., 2016), this delayed effect may suggest that language gains take time to manifest, though they could also reflect natural developmental progression. Conversely, although Let's Play aims to equip caregivers with skills to support enduring child and caregiver outcomes, no further improvements in engagement, behavior, or quality of life were observed post-support, suggesting booster sessions may be needed.

Caregivers experienced significant improvement in parental distress, psychological distress (i.e., symptoms of depression and anxiety), and self-perceived parenting competence from baseline to immediately post-support. Between post-support and follow-up, caregivers' parenting competence continued to improve, underscoring the potential benefits of ongoing, independent application of strategies introduced during the program. These findings highlight the dual benefits of Let's Play for both child and caregiver well-being and support prior research illustrating the critical role of caregiver mental health and well-being in promoting positive caregiver-child interactions and child development (Dykens et al., 2014; Estes et al., 2021; Osborne et al., 2007). Although the directionality of these effects remains unclear, the findings contribute to the limited evidence base on caregiver well-being as a key outcome of early support programs (Nevill et al., 2018; Reichow et al., 2024; Trembath et al., 2023).

The between-group comparisons provided further evidence for program-specific effects. Significant Group × Time effects emerged for parental distress, child behavior, child health-related quality of life, and all secondary caregiver outcomes (depression, anxiety, and parenting competence) when comparing AS with those in the WLC. These effects indicate that improvements were not solely due to passage of time or natural maturation but can be attributed to participation in Let's Play.

Caregivers rated the program as highly acceptable and effective, with strong workshop and coaching attendance (i.e., 64 families attended all three group workshops and 48 attended all five home visits). Participants emphasized increased autism knowledge, enhanced play and communication skills, strengthened caregiver–child relationships, and opportunities for social connection. Interviewed caregivers did not report concerns regarding cultural responsiveness and appreciated the integration of te ao Māori elements such as karakia and the participation of people from a range of cultures, within workshops. However, cultural and intergenerational differences in understanding and acceptance of autism were noted as important to consider. These perceptions are critical to real-world sustainability and support earlier findings on the benefits of group-based coaching in fostering social learning and networks (Jurek et al., 2023; van Noorden et al., 2025) and the often-unseen value of culturally-responsive and inclusive service delivery.

In Aotearoa New Zealand, early support remains difficult to access due to systemic and workforce barriers, especially given rising autism diagnosis rates. Existing models are often therapist-delivered and resource-intensive. The current findings provide preliminary evidence that a brief (9-week), hybrid, community-based approach can effectively support Autistic children and those awaiting formal diagnosis while reducing reliance on specialist-delivered services. Moreover, the qualitative data identifies important components of caregiver coaching and support. For instance, caregiver materials included both theory on what play is and why it is important, as well as practical play skills and examples. Further confirmation and solidification of caregiver understanding was then ensured through in-home coaching. In addition, embedding group coaching within the program allowed for social support networks to be created and enhanced. While further research is needed, it is possible that hybrid models of support may promote appropriate application of knowledge and skills, while creating social supports that can reduce caregiver stress and promote well-being.

The results also highlight the importance of caregiver well-being within early support. Parents of Autistic children often experience elevated stress and mental health challenges (Barroso et al., 2018; Ooi et al., 2016; Zaidman-Zait et al., 2010). While this study cannot isolate the mechanisms underlying caregiver well-being improvements, they may stem from program content, social connection, improved child outcomes, or increased parenting competence. The findings underscore the need for further research into interventions that target caregiver well-being (Barroso et al., 2018; Byrne et al., 2021; Neece et al., 2024).

While the program was designed to address barriers to access (i.e., allowing children on assessment waitlists to access the program), some persisted. Although these barriers were not consistently reported, the need to take time off work, scheduling around family routines and other support appointments, organizing babysitters and travel to attend the community clinic were identified by individuals during the interview process. Potential solutions to such barriers (e.g., digital models) should continue to be explored.

While this research had multiple strengths, the findings should be considered in light of its limitations. First, the lack of a significant Group x Time effect for caregiver-child engagement represents an important limitation. This finding suggests that improvements in engagement over time cannot be confidently attributed to the program itself and

may indeed reflect developmental progression or measurement-related factors. Future research should investigate the role of program dosage, measurement sensitivity, and the potential need for more targeted or sustained strategies to enhance this outcome. Second, only moderate IOA was calculated for the number of utterances coding, suggesting there were differences in how the two SALT-NZAU coders coded utterances. Due to resourcing constraints, there was no single primary language coder for this study (PJ coded 58 videos and JP coded 143; 41 videos used for IOA). It is possible that the absence of a support effect on children's number of utterances may be due to the low IOA, rather than no effect being present. Third, flooding and Cyclone Gabrielle in Auckland during the study affected families' ability to access services and travel around the city, as well as the delivery of assessment packs. Extensions were given to these families to allow them more time to complete their assessments in the face of these unforeseen circumstances, potentially extending the window for natural developmental changes to occur post-support. Finally, the time and disruption component of parent perception of the acceptability of the program was low (14.4 out of a possible 21). This comes from the necessity of needing both multiple group and one-on-one sessions for parents and children. Future research could replicate this study and compare the effectiveness of the program with fewer contact hours to land at the most effective and acceptable version of the program.

Conclusions

Together, the RCT findings suggest multiple positive outcomes from participating in the Let's Play program, including improvement in parental distress, caregiver psychological distress, caregiver self-perceived parenting competence, child health-related quality of life, and child behavior. These support effects were generally significantly greater after participating in Let's Play compared to those in the WLC condition. Caregivers rated the program as highly acceptable and culturally reflexive, and lauded the program for increasing their knowledge and skill, providing opportunities to connect with other families with an Autistic child, and enhancing caregiver-child relationships.

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Declarations

Conflict of interest The authors declare that they have no conflicts of interest relevant to the content of this article.

Ethical Approval Ethical approval was granted by the New Zealand Health and Disability Ethics Committee (Ref #13041).

Informed Consent Informed consent was obtained from all participants included in this study.

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